ITCS1030 CYBER SECURITY PRACTICAL PROJECT (MAJOR LAB PROJECT)

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Student ID:

Introduction

- In this project, I will design a university network with two buildings, mainly divided into main campus and branch campus. The design concept of this project focuses on building a secure and stable network to meet the network needs of a total of 5,000 people on campus. Covering two building on the main campus and the branch campus.

Project Scope

- In my project, the university is mainly divided into main campus and branch campuses. The main campus has 7 floors, and the branch campus has 5 floors. These are approximately 800 faculty and staff and approximately 4,200 students on the main and branch campuses.

- In this project, the technical configuration involved in the two campuses will include DHCP, ACL, NAT, VPN , Routing, Dial peer, Telephony-service and SSH.

- Equipment for both campuses includes routers, multilayer switch, switch, firewall, WLC controllers, access point, and IP phones.

Work Plan

University Network Design	Week1	Week2	Week3	Week4	Week5	Week6	Week7	Week8	Week9	Week10	Week11	Week12	Week13
1. Search for final year project topics													
2. Design university network													
3. Configure VLAN and DHCP													
4. Configure WLC Controller and AAA Server													
5. Configure firewall, ACL, VPN.													
6. Configure Nat on external network.													
7. Testing university network													
8. Writing Final year project documentation.													

Project Resources

- 1. Hardware Resources:
- The hard ware I will be using for this project consists of a laptop.

- 2. Software:
- In this project , I will be using software including Cisco packet tracer, Draw.io, Microsoft Word, and Microsoft PowerPoint, Microsoft Excel , and google chrome software to complete my final project.

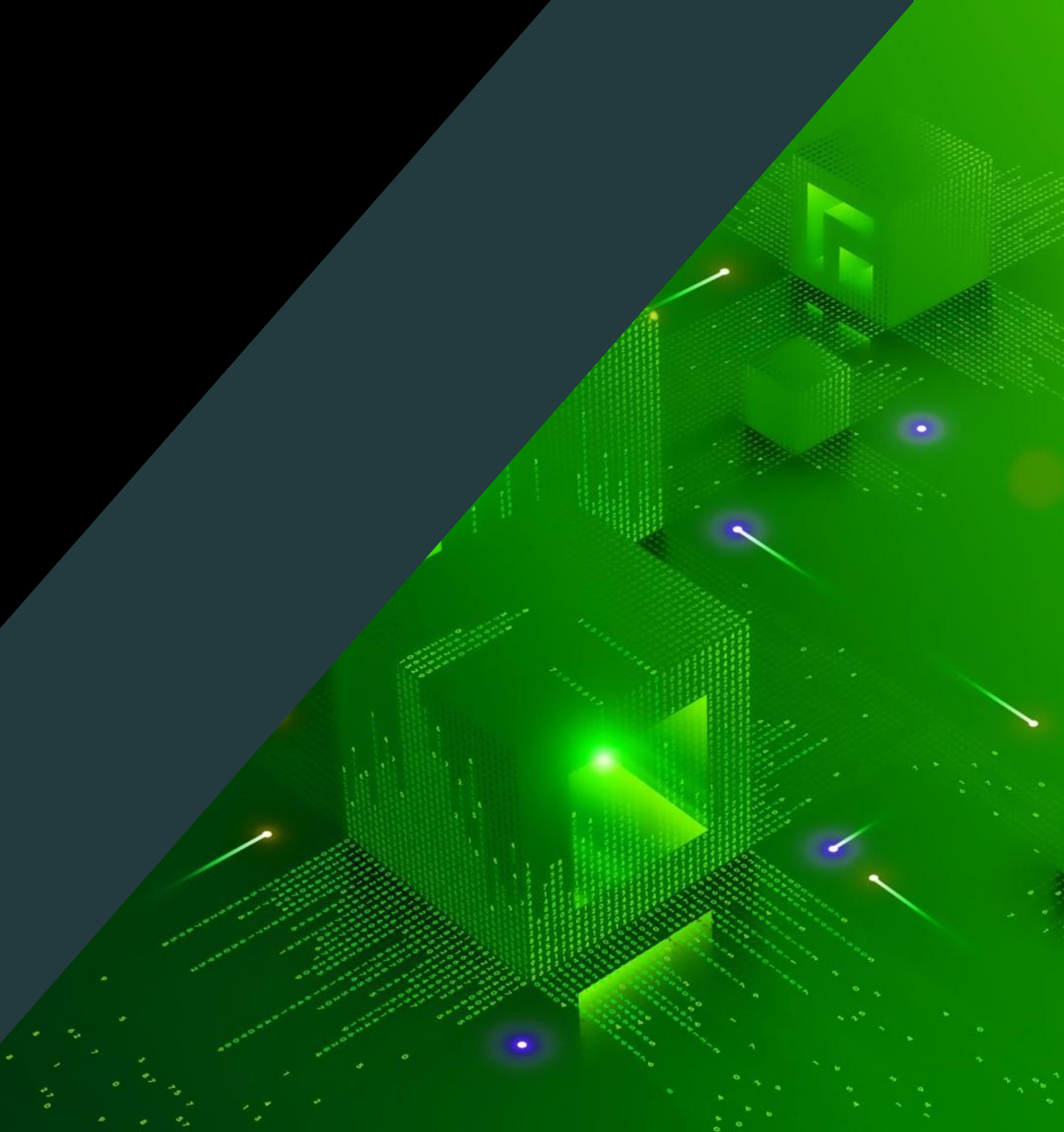
- 3. Access to information:
- Internet – Google.com/ Youtube.com/Git Hub/ Bard and ChatGPT.

- 4.Human Resources:
- Project Supervisor :
- 1 .Mr. Gan Boon Siang
- 2. Mr. Daniel Yap Kim Pun
- User involve: Lee Han Liang

Project Objectives

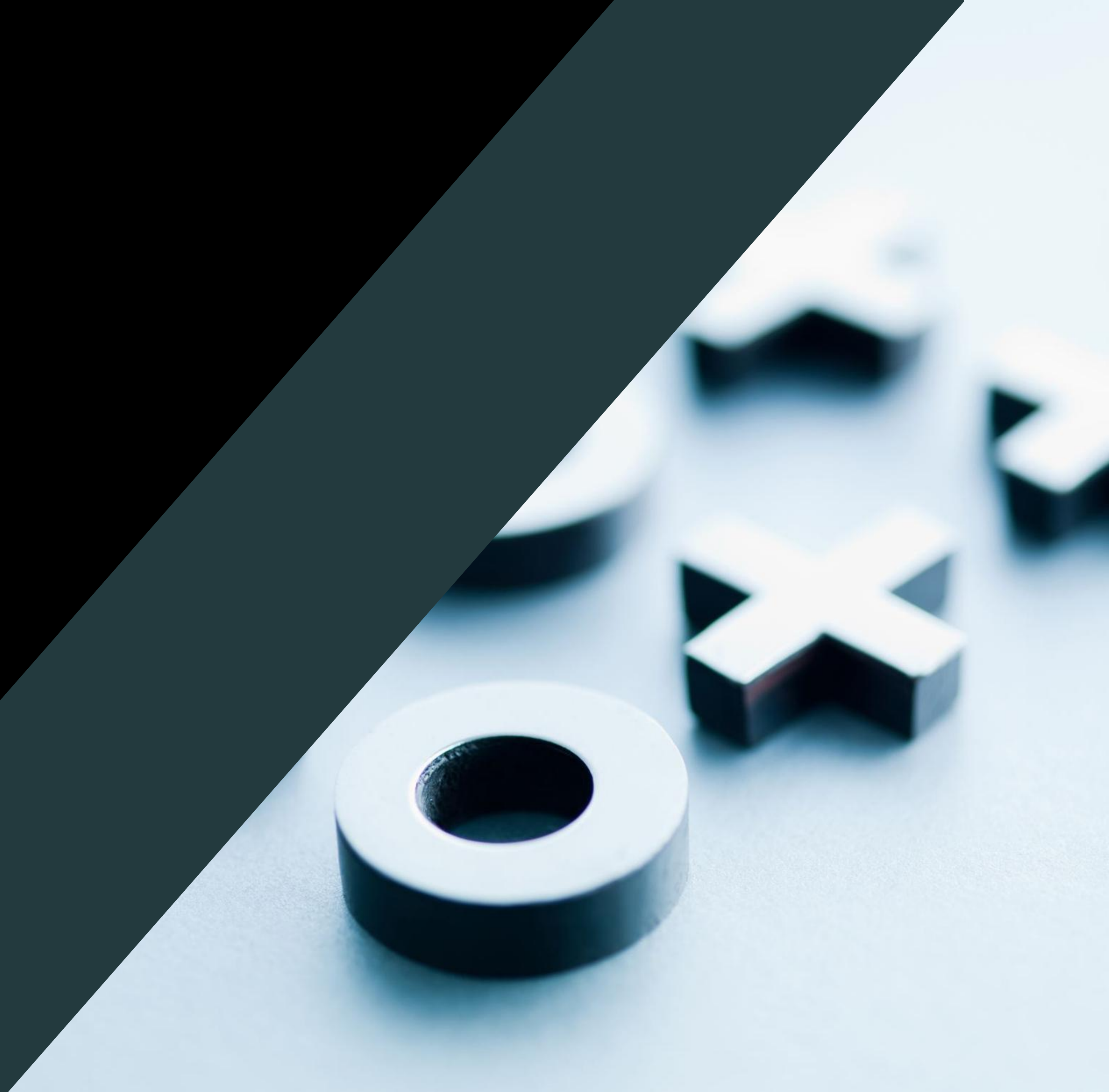
- The project will meet the following objectives:

- DHCP: auto assign IP to all devices.
- ACL: Block unknow network traffic and authorize specific users.
- VLAN: Isolate faculty, staff, students, and guest networks.
- NAT: Convert the real IP of the internal network to the virtual IP of the external network.
- AAA Server: AAA is used to verify the Wi-Fi account password and FTP account password.
- CCTV System: Improve security levels in server rooms and campuses.
- Door Access system: Ensure only authorized personnel have access to restricted areas.
- WLC controller: Used to control access points.
- VPN: Encrypted network connections for main campus and branch campuses.
- Firewall: Provide a more secure network for the intranet.



Conclusion

- I encountered a lot of difficulties in this project and was forced to abandon some of the techniques I wanted to add to the project. I believe that this failure will improve my own abilities. It also reminds me that not everything is smooth sailing and that perseverance is the most important thing. I also believe that these techniques will help me in my future endeavors.



THANK